

- Model Improvement uchun maqsad
- Metrics $\rightarrow$ Advanced Metrics
- Threshold tuning
- Probability of class $\rightarrow y_{pred} = 1$
- Threshold tuning and Advanced Metrics

Model Improvement ^{Rivjantirish} uchun maqsad

- Advanced metrics and decision making
- Model stability
- Advanced Feature Engineering
- Feature Selectionni boshqarish $\Rightarrow$ Dimensional Reduction
- Hyperparameter (tuning) Optimization
- Robustness and Cross Validation Strategy
- Problem solving (guz) • 2 ta Project

Metrics $\Rightarrow$ Advanced Metrics

(Classification $\rightarrow$ Classification Report)

generalization $\rightarrow$ metricesda g-test va y-pred ga qarab bitirish biladi.

Metrics $\rightarrow$ Advanced Metrics : Confusion metrics

- Confusion metrics

- TP $\rightarrow$ Model positive classlarni necha marta to'g'ri predict qilgan (Qurol aniqlandi)
- FP $\rightarrow$ Model positive predict qilgan lekin negative bo'lgan (og'riqlik qaratib hato bo'lgan)
- TN $\rightarrow$ Model negative classni to'g'ri predict qilgan (Kuzgular xatoliklar deb topildi)
- FN $\rightarrow$ Model negative predict qilgan lekin positive bo'lgan (Qurol xatoliklardan o'tib ketib qoldi)

1. Confusion Matrix

Actual Positive

Actual Negative

Predicted Positive

TP (True Positive)

FP (False Positive)

Predicted Negative

FN (False Negative)

TN (True Negative)

2. Accuracy (Modelning umumiy to'g'ri bashorat qilish ulushi)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

3. Precision (Mustah deb topilgan natijalarning qanchasi haqiqatdan ham mustah ekanligini ko'rsatadi)

$$\text{Precision} = \frac{TP}{TP + FP}$$

4. Recall (Sensitivity, True Positive Rate)

(Majlis muvaffaqiyat qoʻlga kiritilgan korraladi.)

$$\text{Recall} = \frac{TP}{TP + FN}$$

5. Specificity (True Negative Rate)

(Majlis muvaffaqiyat qoʻlga kiritilgan korraladi.)

$$\text{Specificity} = \frac{TN}{TN + FP}$$

6. F1-Score (Precision va Recallning oʻrtacha arifmetikasi)

$$F1\text{-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 = \frac{2TP}{2TP + FP + FN}$$

7. False Positive Rate (FPR)

$$FPR = \frac{FP}{FP + TN}$$

8. False Negative Rate (FNR)

$$FNR = \frac{FN}{FN + TP}$$

9. Balanced Accuracy

(Klaslar muvozanatligi boʻlganda ishlatiladi)

$$\text{Balanced Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2}$$

10. Jaccard Index

$$\text{Jaccard} = \frac{TP}{TP + FP + FN}$$

11. Matthews Correlation Coefficient (MCC)

(Klaslar muvozanatligi boʻlganda juda foydali)

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

12. ROC-AUC (ROC egrisi dastgiri oʻrtacha miqdori)

ROC muvaffaqiyatligi

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

AUC yigʻindisi oʻrtacha miqdori

Regression value method.

 $y_i \rightarrow$ target value $\hat{y}_i \rightarrow$ model predicted value $n \rightarrow$ number of data $\bar{y} \rightarrow$ target value

1. MAE (Mean Absolute Error) - Yatawung mullay puyathu orthadasi

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (\text{Qandara wathara shudhara puyathu})$$

2. MSE (Mean Squared Error) - Kutha wathara mullay puyathu

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{Yatawung wathara orthadasi})$$

3. RMSE (Root Mean Squared Error) - MSE mullay wathara mullay

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (\text{Mullay wathara mullay puyathu})$$

4. R^2 Score (Coefficient of Determination) - Model mullay puyathu
significance puyathu puyathu puyathu puyathu

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$R^2 = 1 \rightarrow$ ideal model
 $R^2 = 0 \rightarrow$ orthadasi puyathu orthadasi
 $R^2 < 0 \rightarrow$ puyathu puyathu

Example:

Target value $y = [3, 5, 7, 9]$ Model predicted $\hat{y} = [2.5, 5.5, 6.5, 8.5]$ 1-puyathu: Target value $\bar{y} = \frac{3+5+7+9}{4} = 6$

2-puyathu: SSE (Residual Sum of Squares)

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (3-2.5)^2 + (5-5.5)^2 + (7-6.5)^2 + (9-8.5)^2 = 0.25 + 0.25 + 0.25 + 0.25 = 1$$

3-puyathu: SST (Total Sum of Squares)

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = (3-6)^2 + (5-6)^2 + (7-6)^2 + (9-6)^2 = 9 + 1 + 1 + 9 = 20$$

4-puyathu: R^2 mullay puyathu

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{1}{20} = 0.95$$

Xulosa:

- Model 98% accuracy, however human diagnosis model is less accurate.
- You want 98% like model, you want your diagnosis accurate.

Threshold tuning

- Threshold tuning = decision controller
- Threshold = shifting the balance between model's yes and model's no
- You want to shift the balance

Analogy: moving goalposts when threshold is low

- Business goal
- Risk tolerance
- Cost of errors (FP vs FN)
- Class imbalance

Probability of class

Classification

Prediction

0 = Negative

1 = Positive

Model $y = \text{pred}[0.3] \Rightarrow 1$ (0 you are 30% likely to be 1)

probs = model.predict_proba(X_test)

print("Raw probabilities", probs[0])

positive - prob = probs[0][1]

print("1st model confidence for positive class: 1 - positive - prob = 100% - 24.3%")

Threshold: (probability $\geq$ threshold)

- Random guess: 50% chance
- Threshold: 0.5
- Low (0.3)
- Medium (0.5)
- High (0.8)

Threshold level: Best Use Case

Low threshold

When missing positives is dangerous (need high recall)

Medium threshold

Default for balanced classes with equal error cost

High threshold

When false positives are dangerous (need high precision)

Analogy

Fire alarm: better to over-alert than miss a real fire

Normal exam pass mark: fair, not too strict or easy

Border confirming cancer: avoid false alarms

Printing:

import numpy as np
probs = np.array([

[0.10, 0.90],

[0.30, 0.70],

[0.40, 0.60],

[0.70, 0.30],

[0.80, 0.20],])

⇒

for i, p in enumerate(probs):

print(f"Sample {i+1} Low probability: ", p)

positive_prob = p [1]

print(f"Model confidence: {positive_prob * 100:2.5f}%")

print()

Three sholdan $\rightarrow$ high, low, & medium

import numpy as np

from sklearn.metrics import precision_score, recall_score

Example model probabilities

probs = np.array([0.90, 0.70, 0.60, 0.30, 0.20])

True labels

y_true = np.array([1, 1, 0, 0, 1])

print("Predicted Probabilities:", probs)

print("True Labels:", y_true)

1. Low Threshold (0.3) $\Rightarrow$ Recall - high

low_thr = 0.3

y_pred_low = (probs >= low_thr).astype(int)

print("m == Low Threshold (0.3) ==>")

print("Predictions:", y_pred_low)

print("Precision:", precision_score(y_true, y_pred_low))

print("Recall:", recall_score(y_true, y_pred_low))

2. Medium Threshold (0.5)

med_thr = 0.5

y_pred_med = (probs >= med_thr).astype(int)

print("m == Low Threshold (0.5) ==>")

print("Predictions:", y_pred_med)

print("Precision:", precision_score(y_true, y_pred_med))

print("Recall:", recall_score(y_true, y_pred_med))